. The circle $C$ has equation

$$x^2 + y^2 + 4x - 30y + 209 = 0$$

(a) Find

(i) the coordinates of the centre of $C$,

(ii) the exact value of the radius of $C$.

**(3)**

The line $L$ has equation $y = mx + 1$, where $m$ is a constant.

Given that $L$ is the tangent to $C$ at the point $P$,

(b) show that

$$2m^2 - 7m - 22 = 0$$

**(3)**

(c) Hence find the possible pairs of coordinates of $P$.

**(4)**